

Lecture 1: Basic Concepts

EE201 – Circuit Theory I

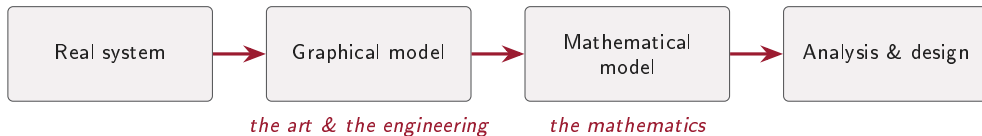
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charge ■ current ■ voltage ■ power ■

these colours mean the same thing on every slide and in the notes

Circuit theory is not about wires



- A real resistor has inductance, a temperature coefficient, a size, a price.
- We keep **one number**, R , and throw the rest away.
- Whether that is brilliant or foolish depends on the question you are asking.

“All models are wrong, but some are useful.”

George E. P. Box, 1976

The same pipeline, other domains

Domain	“Across” quantity	“Through” quantity
electrical	electric potential	current
hydraulic	pressure	volumetric flow
thermal	temperature	heat flow
mechanical	velocity	force

“Across” needs two points — “through” needs one

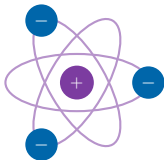
An across quantity is a **difference**, and the word is usually dropped: sometimes for convenience (both ends of R_2 are obvious), more often because a reference was agreed on long ago — altitude from sea level, gauge pressure from atmospheric pressure. A circuit has no universal reference, so we pick one and call it *ground*.

A through quantity is measured **at** one place and needs a *direction*, not a second point — and no reference at all: zero current means the same to everyone.

Charge

Definition

Charge is the property of matter that makes it feel a force in an electromagnetic field. Symbol q , unit **coulomb (C)**.



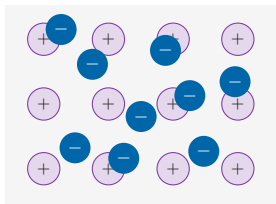
one atom: as many $-$ as $+$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ C} \approx 6.24 \times 10^{18} \text{ electron charges}$$

- electron: $-e$ proton: $+e$
- **Charge is conserved** — never created or destroyed, only moved. (This will become KCL.)

A wire is already full of charge



copper: fixed + ions,
electrons free to wander

1 cm³ of copper contains about

8.5×10^{22} free electrons $\approx$ 13 600 C

of *mobile* charge — and its net charge is *zero*.

So what is a current?

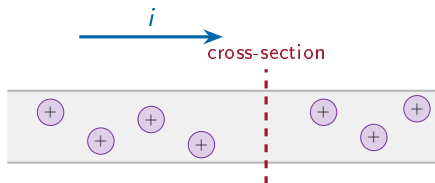
Every free electron left a + ion behind. The metal stays neutral; current is that charge *drifting*, not charge being poured in.

Current: charge in motion

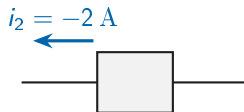
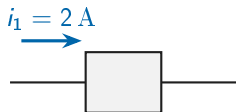
Definition

$$i(t) = \frac{dq(t)}{dt} \quad \Longleftrightarrow \quad q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau$$

Unit: **ampere (A)**, $1 \text{ A} = 1 \text{ C/s}$.



The arrow is a choice, not a fact



Same physical situation. The arrow says which way we call positive; the sign of the number says which way the charge really goes.

A negative answer is not a mistake — it is the model telling you something.

Two numbers worth knowing

How much is a coulomb?

A 4000 mAh phone battery:

$$Q = 4 \text{ A} \times 3600 \text{ s} = 14\,400 \text{ C}$$

$$\approx 9 \times 10^{22} \text{ elementary charges}$$

At 10^{22} charges, the lumpiness of charge is invisible. We may treat $q(t)$ as a **smooth** quantity — which is what lets us write dq/dt at all.

How fast do the charges move?

1 A in a 1 mm^2 copper wire:

each electron drifts at about

$$0.07 \text{ mm/s} \approx 1 \text{ metre in 4 hours}$$

Yet the lamp lights the instant you flip the switch. The carriers crawl; the *signal* that starts them travels near the speed of light.

We come back to that gap when we define a *lumped* circuit.

Conventional current

The arrow points the way *positive* charge would move

In a copper wire the carriers are electrons, so they actually move *against* the arrow.

No equation in this course changes because of it.

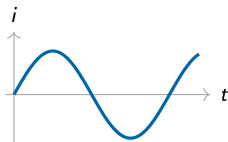
A current counts only the **net rate at which charge crosses a section**: one electron moving left transports exactly as much charge to the right as one positive charge moving right. The model never asks what the carriers are — and it had better not, because in a semiconductor they are electrons *and* holes, and in a battery they are ions.

Notation

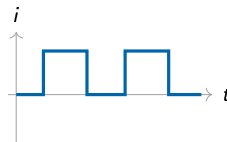
may vary in time constant
 $i(t), v(t), q(t), p(t)$ I, V, Q, P



constant (DC)



sinusoidal (AC)



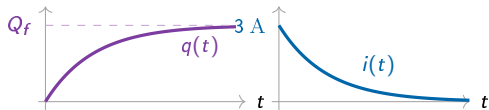
pulsed

Only the first deserves the symbol I .

Example: charging a phone

The last part of a charge cycle: the charge *stored in the battery* approaches its full value, $q(t) = Q_f (1 - e^{-t/\tau})$, with $Q_f = 3600 \text{ C}$ and $\tau = 20 \text{ min}$.

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau} e^{-t/\tau}$$

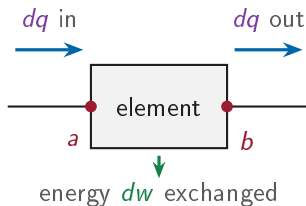


- $i(0) = 3600/1200 = 3 \text{ A}$
- after 20 min: $i = 3/e \approx 1.1 \text{ A}$
- the fuller it gets, the slower it fills

Charge is what accumulates; current is how fast it accumulates.

Voltage: why the charge moves

Charge does not move by itself. Pushing it through an element **costs or releases energy** — and that is what voltage measures.



Definition

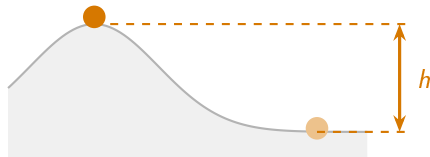
$$v_{ab} = \frac{dw}{dq} \quad \text{unit: } \mathbf{volt\ (V)}, \quad 1\ \text{V} = 1\ \text{J/C}$$

the energy exchanged *per unit of charge* carried from a to b through the element.

What that definition does — and does not — say

- It is a **ratio**. Carry twice the charge and twice the energy is exchanged, so v_{ab} comes out the same. Voltage describes the **pair of points**, not the charge that passed between them.
- It is defined even when **nothing is moving**. A 1.5 V battery on the shelf has 1.5 V across it with $i = 0$ — just as a closed tap still has pressure behind it.
- Order matters: $v_{ab} = -v_{ba}$. There is no “voltage at a point” until we fix a reference (Section: ground).
- In a *lumped* circuit the energy per charge depends only on the two endpoints, **not on the path** taken between them. That is what makes “the voltage between a and b ” a meaningful phrase — and it becomes KVL.

Voltage is an electrical height



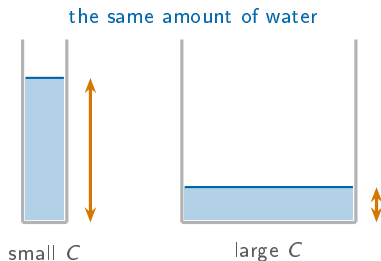
energy per unit *mass* to lift: gh

- Only the height **difference** matters.
- “Sea level” is a **choice** — and changing it changes no physics.
- A heavier ball gains more energy, but energy *per kilogram* is the same: that is why gh , like voltage, is a property of the two places.

Voltage is the electrical version of this, with charge in place of mass.

Charge is the amount, voltage is the level

$$q = C v$$

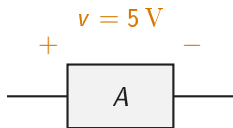


- **charge** — the amount poured in
- **voltage** — the level it reaches
- C — the cross-section of the container

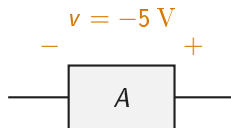
The **shape of the circuit** plays the part of the shape of the tank: the same charge poured between *another* pair of points reaches a different level, because C is different. We define C properly in Lecture 5.

Reference polarity

Exactly as for current, we mark $+$ and $-$ **before** we know the answer.



left terminal 5 V above the right one



same element, *same* state

Same physical situation, two bookkeeping choices — exactly like the current arrow.

A first element: the resistor

We have defined the variables. To talk about a circuit we also need an *element* — a rule tying v and i together. Here is the simplest one.

Ohm's law

$$v = R i$$

R in ohms (Ω), $1 \Omega = 1 \text{ V} / \text{A}$



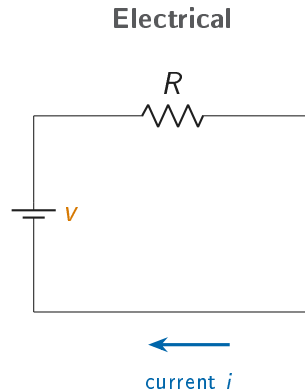
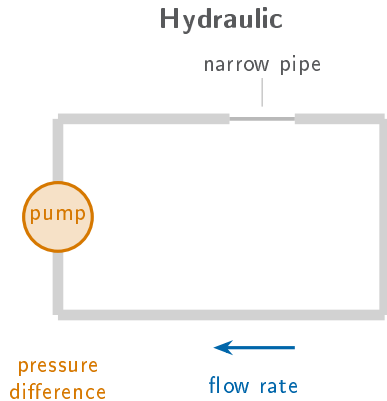
- **Algebraic, no memory:** the current *now* depends only on the voltage *now* — no derivatives, no history.
- **Geometry sets R :** for a uniform bar,

$$R = \rho \frac{L}{A}$$

long and thin $\Rightarrow$ large R ; short and fat $\Rightarrow$ small R . Exactly like a pipe.

The full family of elements — and the sign convention that goes with them — comes in Lecture 2.

The hydraulic analogy



Across and through

Hydraulic	Electrical	
pressure (difference)	voltage v	across: a difference between two points
volumetric flow rate	current i	through: needs a direction
pipe restriction	resistance R	opposes flow, dissipates energy
pump	voltage source	supplies energy

Ground: the node we choose to call zero

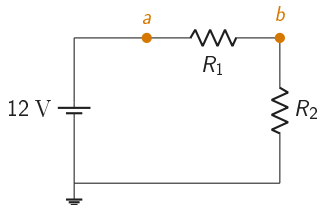
Reference node

Only differences matter, so we are free to declare one node to be 0 V. Then

$$V_{ab} = V_a - V_b.$$

Move the reference and *every* node voltage shifts by the same constant. No two-point voltage, current or power changes.

Pick the node that the most elements touch — your equations get shorter.



Three different things are called “ground”

Earth ground

A physical connection to the planet. Safety.

Chassis ground

The metal enclosure of the instrument.

Signal ground

The analysis reference. **This is ours.**

In EE201 we always mean the third one.

In the laboratory, confusing them is how you blow a fuse.

Power falls out of the definitions

$$p(t) = \frac{dw}{dt} = \underbrace{\frac{dw}{dq}}_{v(t)} \cdot \underbrace{\frac{dq}{dt}}_{i(t)} = v(t) i(t)$$

Power

unit **watt (W)**, $1 \text{ W} = 1 \text{ J/s}$

Energy

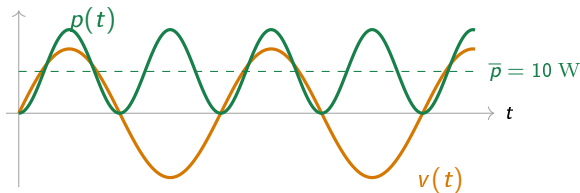
$$w = \int_{t_0}^t v(\tau) i(\tau) d\tau$$

$p = vi$ was **not postulated** — it is a consequence of defining v as energy per charge and i as charge per time.

Power that changes in time

A resistive element with $v(t) = 10 \cos \omega t$ V and $i(t) = 2 \cos \omega t$ A:

$$p(t) = vi = 20 \cos^2 \omega t = 10(1 + \cos 2\omega t) \text{ W}$$



- $p(t) \geq 0$: a resistor never gives energy back.
- The power ripples at **twice** the frequency, around $\bar{p} = 10 \text{ W}$.

The passive sign convention

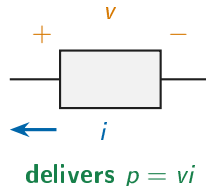
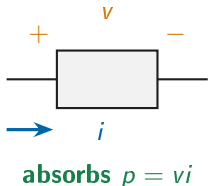
PSC

Draw the current arrow so that it **enters the + terminal**. Then

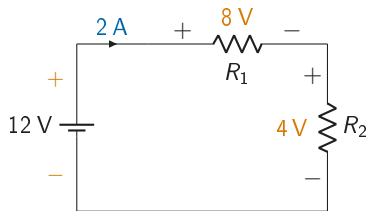
$$p = vi = \text{power absorbed by the element.}$$

$p > 0$: the element absorbs. $p < 0$: it delivers.

If the picture you are given does *not* obey the PSC, do not memorise a second formula: flip one reference, change that variable's sign, and use $p = vi$ as always.



Example: a power balance



$$p_{R_1} = 8 \times 2 = +16 \text{ W}$$

$$p_{R_2} = 4 \times 2 = +8 \text{ W}$$

$$p_{\text{src}} = 12 \times (-2) = -24 \text{ W}$$

$$16 + 8 - 24 = 0 \checkmark$$

The source current *leaves* its + terminal —
hence -2 A .

In *any* circuit, at *every* instant, $\sum_k p_k(t) = 0$. If the powers in your solution do not cancel, your solution is wrong — **a free check, every time.**

What exactly is a circuit?

An **electric circuit** is an interconnection of elements through which charge can flow along closed paths. Everything in EE201 also assumes it is **lumped**:

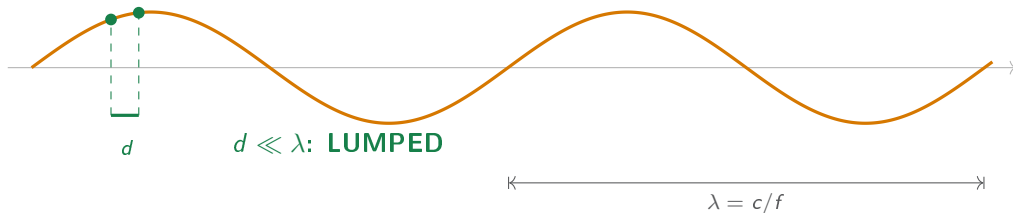
Three things we assume

- 1 Nothing piles up.** The current leaving an element equals the current entering it, at every instant.
Charge may *separate* inside — that is what a capacitor does — but the element as a whole never gains or loses charge.
- 2 Nothing leaks out.** No changing *magnetic* field passes through the loops of your wiring: they are neither antennas nor transformers.
- 3 Nothing is late.** The circuit is much smaller than the wavelength of the signal, $d \ll \lambda = c/f$ (c = speed of light).

50 Hz: $\lambda = 6000$ km, any board is lumped. 2.4 GHz: $\lambda = 12.5$ cm, a 3 cm track is not.

Nothing is late: a circuit much smaller than λ

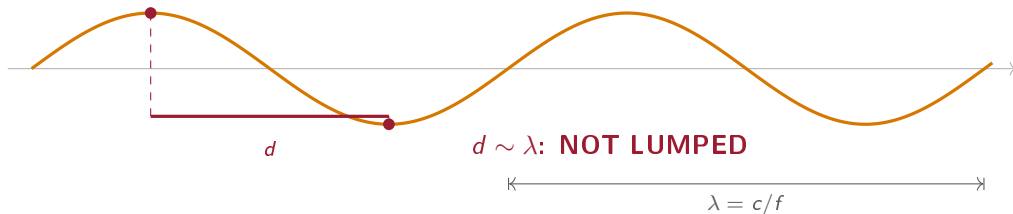
a *snapshot* along the circuit at one instant — the axis is distance, not time



Both ends of the circuit sit at practically the same point of the wave. Every point of the circuit reads **the same number** at any instant — so “the voltage of a node” means something, and one time variable t describes the whole circuit.

... and one that is not

a *snapshot* along the circuit at one instant — the axis is distance, not time

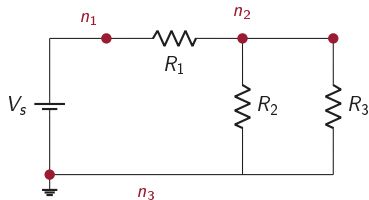


One end is at the crest while the other is at the trough. Different parts of the same circuit hold **different values at the same moment** — there is no single number to call the node voltage, and circuit theory no longer applies.

Terminals, nodes, branches, loops

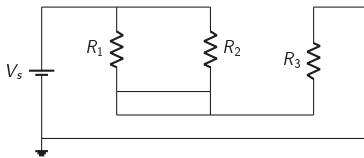
- **Terminal** — a point where an element connects. R, L, C: two. Transistor: three.
- **Node** — a connection point. Everything joined by ideal wire is **one** node, however long the wire.
- **Branch** — one element between two nodes.
- **Loop** — a closed path through no node twice.

$$\# \text{branches} = \# \text{loops} + \# \text{nodes} - 1$$

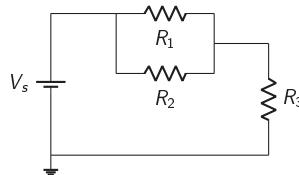


$$4 = 2 + 3 - 1$$

The same circuit, drawn twice



drawing (a)



drawing (b)

Same circuit. **Drawing is not topology.**

When a problem looks impossible, redraw it so that everything joined to the same node sits together — the series and parallel connections then show themselves.

Summary

Quantity	Symbol	Definition	Unit
charge	q	—	C
current	i	$i = dq/dt$	A = C/s
voltage	v	$v = dw/dq$	V = J/C
power	p	$p = vi$	W = J/s
energy	w	$w = \int p dt$	J

- References (arrow, polarity) are **choices**; signs carry the physics.
- PSC: current into $+$ $\Rightarrow p = vi$ is absorbed, and $\sum_k p_k = 0$.
- Across vs through. Ground is a choice.
- Lumped $\Rightarrow$ ODEs. That is what makes this course possible.

Next lecture

Interconnection laws

Kirchhoff's current law (conservation of charge) and Kirchhoff's voltage law (conservation of energy), then the branch relations of the basic elements.

Questions?

Notes for this lecture: see the course repository.